

Solution Architecture

Date	10 July 2026
Team ID	AIML-HDI-2026
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Solution Architecture Diagram

A high-level visual representation of the solution's architecture, mapping the Presentation Layer, Application Layer, and Data Layer for the HDI Intelligence Engine.

Figure 4.1 — Solution Architecture

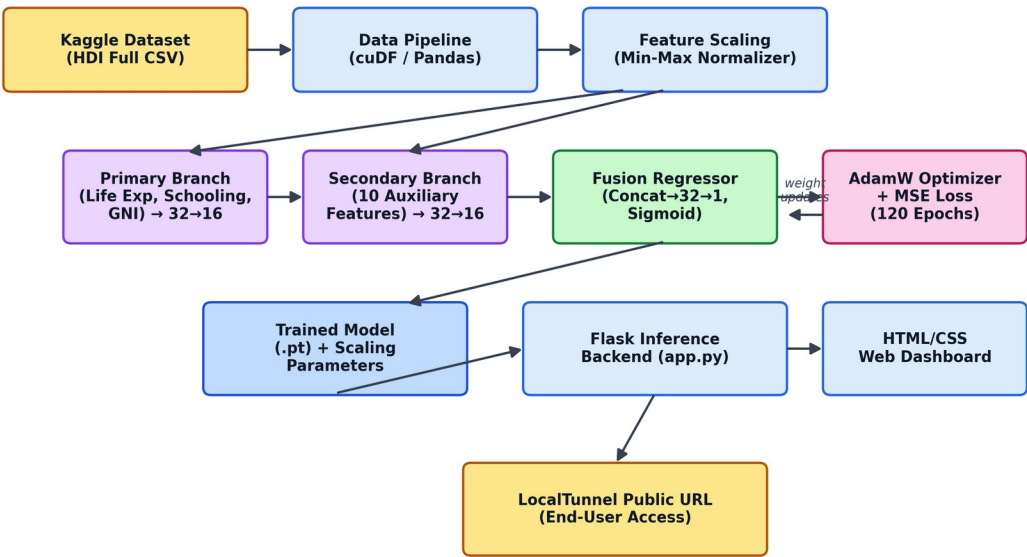


Figure: End-to-end solution architecture, from Kaggle dataset ingestion to the deployed web dashboard.

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for inputting socio-economic indicator values and displaying the forecasted HDI results.	HTML5, CSS3, Jinja2
API Gateway	Acts as the entry point for the application, routing user requests to the backend prediction logic.	Flask (Python), LocalTunnel
Core Logic Service	Handles feature scaling and executes the dual-branch hybrid neural network to perform the	PyTorch (EnhancedHybridHDI Net)

Component Name	Description / Role in Architecture	Technologies Used
	HDI calculation.	
Database / Storage	Stores the trained model weights and Min-Max scaling parameters, and processes incoming user data requests.	PyTorch serialization (torch.save)